Solution Code - PL SQL programming - Triggers

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: solution.sql

```sql
-- Scenario 1: UpdateCustomerLastModified Trigger
CREATE OR REPLACE TRIGGER UpdateCustomerLastModified
BEFORE UPDATE ON Customers
FOR EACH ROW
BEGIN
    :new.LastModified := SYSDATE;
END;
/

-- Scenario 2: LogTransaction Trigger
CREATE TABLE AuditLog (
    LogID NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    TransactionID NUMBER,
    AccountID NUMBER,
    Amount NUMBER,
    LogDate DATE
);

CREATE OR REPLACE TRIGGER LogTransaction
AFTER INSERT ON Transactions
FOR EACH ROW
BEGIN
    INSERT INTO AuditLog (TransactionID, AccountID, Amount, LogDate)
    VALUES (:new.TransactionID, :new.AccountID, :new.Amount, SYSDATE);
END;
/

-- Scenario 3: CheckTransactionRules Trigger
CREATE OR REPLACE TRIGGER CheckTransactionRules
BEFORE INSERT ON Transactions
FOR EACH ROW
DECLARE
    v_bal NUMBER;
BEGIN
    IF :new.TransactionType = 'Withdrawal' THEN
        SELECT Balance INTO v_bal FROM Accounts WHERE AccountID = :new.AccountID;
        IF :new.Amount > v_bal THEN
            RAISE_APPLICATION_ERROR(-20001, 'Insufficient funds for withdrawal.');
        END IF;
    ELSIF :new.TransactionType = 'Deposit' AND :new.Amount <= 0 THEN
        RAISE_APPLICATION_ERROR(-20002, 'Deposit amount must be positive.');
    END IF;
END;
```

```
/
```